

Multilevel Analysis

KU Leuven

Alexander Schmidt-Catran

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Agenda I

This course consists of 12 sessions. In sessions 1-9, we will alternate between lectures and hands-on exercises in R. The last three sessions are dedicated to open questions and student presentations.

Sessions:

- 03/05, 17h-21h: Introduction to hierarchical data structures
- 03/06, 17h-21h: The linear two-level model with random intercepts
- 03/07, 09h-13h: The linear two-level model with random intercepts and random slopes
- 03/12, 17h-21h: Cross-classified, multiple membership and generalized mixed models

Agenda II

Sessions (continued):

- 03/13, 17h-21h: Analyzing panel data
- 03/14, 09h-13h: The within-between specification
- 03/19, 17h-21h: Contextual effects in multilevel models
- 03/20, 17h-21h: Multilevel models with more than two levels, the within-between model for pooledcross-sectional data
- 03/21, 09h-13h: Comparing and testing in multilevel modeling, estimation techniques, advanced topics
- 03/26, 17h-21h: Open session: Q&A, talking about course assignments
- 03/27, 17h-21h: Presentation of research proposals from students
- 03/28, 09h-13h: Presentation of research proposals from students

We are using R for this class.

- You should have RStudio installed on your computer.
- Download and extract the file `KUL_MultilevelAnalysis24.zip` from the Toledo course.
 - all data and code are included in this folder
 - run the file `RunFirstToPrepareRAndData.R` first, it should set up R for the hands-on exercises
 - this document is written in R Markdown and also included in the zip-file

Session 4: Cross-classified, multiple membership and generalized linear models

Agenda Session 4

Today:

- Reminder: What have we learned so far?
- How to build your models?
- Cross-classified models
- Exercise 7: Estimating cross-classified and multiple membership models in R
- Multiple membership models
- Generalized mixed models
- Exercise 8: Estimating generalized mixed models in R

Reminder: What Have We Learned So Far?

- Multilevel models allow to...
 - account for complex data structures
 - account for statistical dependence in the data
 - model heterogeneity (in levels and effects)
 - model contextual effects
- So far we have only talked about simple linear two-level models for data with strict hierarchical nesting

Reminder: The Random Intercept Model

The two-level random intercept model:

$$y_{ij} = \beta_0 + \beta_1 x_{1ij} + \gamma_1 z_{1j} + u_j + e_{ij}$$

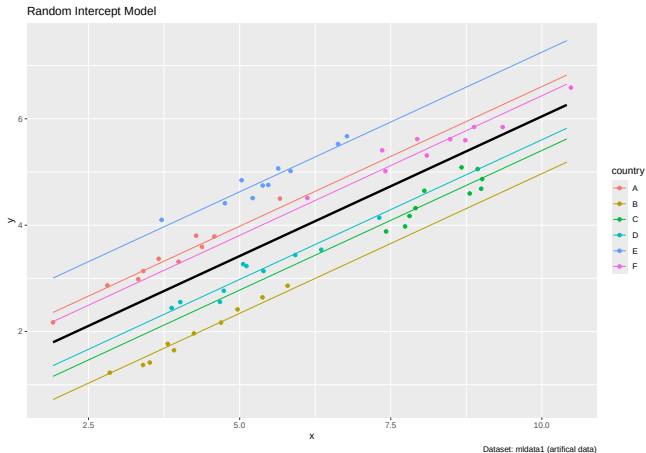
- The model estimates the overall intercept β_0 and the cluster-specific deviations from it (u_j).
- Coefficients of x_{1ij} and z_{1j} are fixed
- Assumptions about the error term:
 - Both error terms are IID (independent and identically distributed):

$$u_j \sim N(0, \sigma_{u0}^2) \text{ and } e_{ij} \sim N(0, \sigma_{e0}^2)$$

- Both error terms are independent of each other:

$$\text{Cov}(u_j, e_{ij}) = 0$$

Reminder: The Random Intercept Model



Reminder: The ICC/VPC

The Intra-class Correlation Coefficient (ICC) or Variance Partition Coefficient (VPC) measures the proportion of variance that is due to differences between clusters:

$$\rho = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_e^2}$$

- It measures the dependency in the data due to the clustering in level-2 units
- Rule of thumb of 10% variation at level-2 means...
 - data is sufficiently non-independent to require adjusted standard errors, i.e. multilevel models
 - there is something substantial to be explained between level-2 units

Reminder: What Can We Do With the RI Model?

- We differentiate between level-1 and level-2 variables (x_{ij} and z_j)
 - level-1 variables can explain both, within- and between-cluster variation
 - level-2 variables cannot explain level-1 variation, but only level-2 variation
- Calculate the reduction of unexplained variances at both levels to evaluate what your model explains
 - use empty model to provide total explained variance
 - compare nested models with each other to evaluate explanatory power of specific variable(s)
- Start ...
 - with an empty model, then
 - introduce level-1 variables next, then
 - add level-2 variables

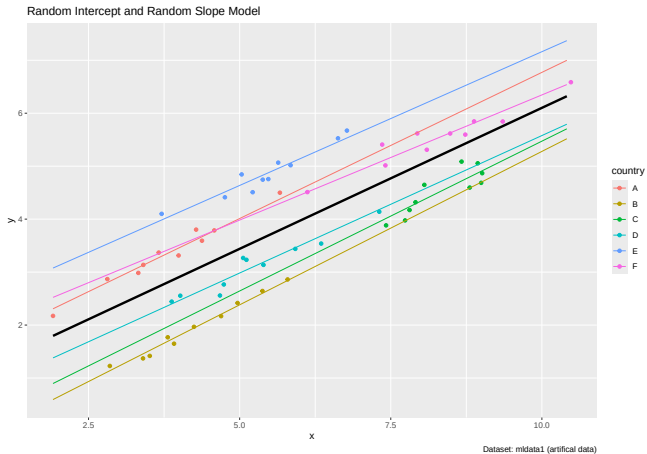
Reminder: The RIRS Model

The random intercept and random slope model:

$$y_{ij} = \beta_0 + \beta_1 x_{1ij} + u_{0j} + u_{1j} x_{1ij} + e_{ij}$$

- Model allows for effect heterogeneity of x :
 $\beta_1 x_{1ij} + u_{1j} x_{1ij} = (\beta_1 + u_{1j}) x_{1ij}$
- Model explicitly accounts for heteroscedasticity, i.e. the error term of the model is heteroscedastic: $u_{1j} x_{1ij}$

Reminder: The RIRS Model



Reminder: The RIRS Model

The error term at level-2 takes the form of a variance-covariance matrix:

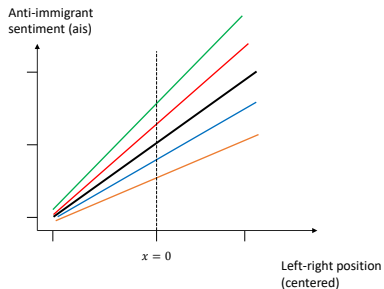
$$\Sigma = \begin{bmatrix} \sigma_{u0}^2 & & & \\ \sigma_{u0u1} & \sigma_{u1}^2 & & \\ \vdots & & \ddots & \\ \sigma_{u0uk} & \sigma_{u1uk} & \dots & \sigma_{uk}^2 \end{bmatrix}$$

- Σ is a symmetric matrix with k rows/columns, where k is the number of random slopes
- The off-diagonal provides the variances of the random effects ($u_{0j}, u_{1j}, \dots, u_{kj}$)
- The covariances between all random effects are estimated parameters (note: R provides them as correlations)

Reminder: The RIRS Model:

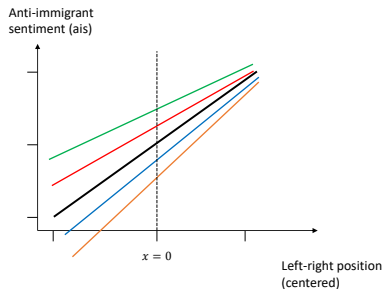
Covariance between intercept and slope is informative about the pattern of fitted lines

'Fanning out' pattern:



$$\sigma_{u_0 u_1} > 0$$

'Fanning in' pattern:



$$\sigma_{u_0 u_1} < 0$$

Reminder: The Importance of Centering

- In multilevel models with random slopes it is important to center the predictor variable(s)
 - variance of random intercept is 'measured' at $x = 0$
 - covariance of intercept and slopes is 'measured' at $x = 0$
 - estimate of covariance can be misleading, when x is not centered

Reminder: Cross-level Interactions in the RIRS Model

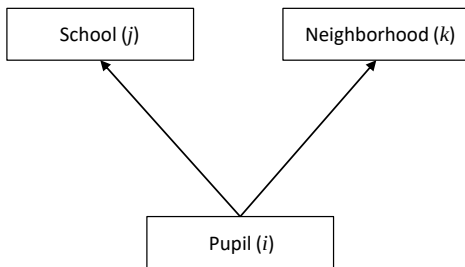
- Cross-level interactions (interactions between x_{ij} and z_j) can explain the variation in slopes
 - calculate variance reduction in slope to get an idea of the explanatory power of such an interaction
 - cross-level interaction always require a random slope of the level-1 variable

How To Build Your Models?

- Always start with an empty/null model and decompose the total variance (How much variation is there at each level?)
- Introduce lower-level variables first to account for compositional effects (How much variation between level-2 clusters is explained by the distribution of level-1 characteristics?)
- Introduce random slopes to account for effect heterogeneity and model heteroscedasticity
- Introduce higher-level variables to test their effects (How much of the remaining variation at level-2 can be explained by level-2 characteristics?)
- Introduce cross-level interaction effects to explain effect heterogeneity at level-1 with level-2 characteristics (How much of the variation in slopes can be explained by level-2 variables?)

Cross-classified Models

Remember our classification-diagrams:



- level-1 (i) units are nested in two (or more) higher level clusters of **different types** (j and k)

The two-way cross-classified model:

$$y_{ijk} = \beta_0 + \beta_1 x_{ijk} + v_k + u_j + e_{ijk}$$

- Model has random intercepts for clusters k and for clusters j
- Assumptions about the error term:
 - All error terms are IID (independent and identically distributed):

$$v_k \sim N(0, \sigma_v^2), u_j \sim N(0, \sigma_u^2) \text{ and } e_{ij} \sim N(0, \sigma_e^2)$$

- All error terms are independent of each other:

$$\text{Cov}(v_k, u_j) = \text{Cov}(v_k, e_{ij}) = \text{Cov}(u_j, e_{ij}) = 0$$

Notation in Cross-classified Models

Notation gets tricky

- $y_{ijk} = \beta_0 + \beta_1 x_{ijk} + v_k + u_j + e_{ijk}$ may be misleading, as ijk normally implies that i is nested in j which is nested in k
- Alternative notations available, e.g.
 - Rasbash and Browne (2001) introduced the multiple subscripts notation:

$$y_{i(j_1, j_2)} = \beta_0 + \beta_1 x_{i(j_1, j_2)} + u_{j_1} + u_{j_2} + e_{i(j_1, j_2)}$$

- Browne, Goldstein, and Rasbash (2001) introduced the classification notation:

$$y_i = \beta_0 + \beta_1 x_i + u_{\text{neighb}(i)}^3 + u_{\text{school}(i)}^2 + e_i$$

- But we stick with the notation from the previous slide

Specifying Cross-classified Models in R

The lmer-function allows for multiple RE matrices:

```
lmer(formula = y ~FES + (REs|cluster) +  
(REs|cluster), data = mydata)
```

Cross-classified models are easily specified. For example:

```
lmer(formula = score ~female + (1|neighborhood) +  
(1|school), data = ssls)
```

- random slopes are specified as before and can be included in both RE matrices
- example: (1 + female |neighborhood) + (1 + female |school), data = ssls)

R Example: Modelling Pig Weight

```
# A cross-classified empty model of pig weight
```

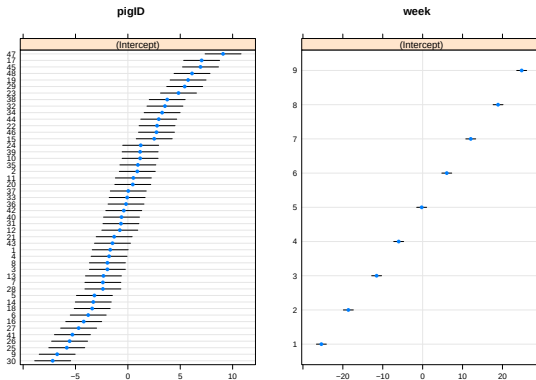
```
pigmodel <- lmer(formula=weight ~ 1 + (1|pigID) + (1|week), data=pigs)
summary(pigmodel)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: weight ~ 1 + (1 | pigID) + (1 | week)
## Data: pigs
##
## REML criterion at convergence: 2086.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.7639 -0.5170 -0.0264  0.4879  3.9911
##
## Random effects:
## Groups Name Variance Std.Dev.
## pigID (Intercept) 15.153 3.893
## week (Intercept) 289.308 17.009
## Residual 4.297 2.073
## Number of obs: 432, groups: pigID, 48; week, 9
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 50.405 5.698 8.158 8.846 1.86e-05 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

R Example: Plotting Two Random Intercepts

Plotting REs from cross-classified model (grid.arrange is part of the gridExtra package):

```
vu <- ranef(pigmodel)      # extract random effects
p  <- dotplot(vu)[["pigID"]] # make plot of pig intercepts
w  <- dotplot(vu)[["week"]]  # make plot of week intercepts
grid.arrange(p, w, nrow=1)  # combine into one plot
```



VPC and ICC in Cross-classified Models

- Total variance has three independent components: σ_u^2 , σ_v^2 , σ_e^2
- Variance partitioning can be done in various ways:
 - How much variance between clusters j :

$$\frac{\sigma_u^2}{\sigma_u^2 + \sigma_v^2 + \sigma_e^2}$$

- How much variance between clusters k :

$$\frac{\sigma_v^2}{\sigma_u^2 + \sigma_v^2 + \sigma_e^2}$$

- How much variance at higher levels (clusters j and k):

$$\frac{\sigma_u^2 + \sigma_k^2}{\sigma_u^2 + \sigma_v^2 + \sigma_e^2}$$

- How much of higher-level variance between clusters j or k :

$$\frac{\sigma_u^2}{\sigma_u^2 + \sigma_k^2} \text{ or } \frac{\sigma_k^2}{\sigma_u^2 + \sigma_k^2}$$

Adding A Random Interaction Effect

- The two-way cross-classified model assumes that the higher-level clusters (k and j) have additive effects
- We can introduce an interaction between the two random effects:

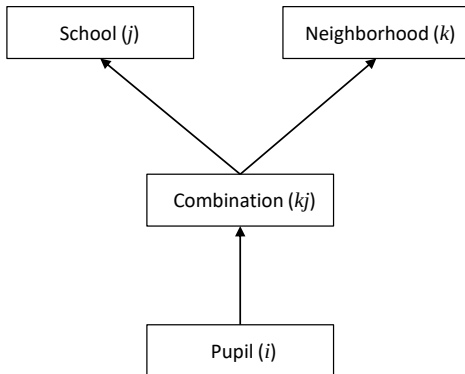
$$y_{ijk} = \beta_0 + \beta_1 x_{ijk} + v_k + u_j + w_{kj} + e_{ijk}$$

- in addition to the additive cluster effects ($v_k + u_j$), we model the effect of specific combinations of clusters k and j : w_{kj}
- only possible if there are multiple observations per combination (not necessarily for every combination)
- However, sometimes these models are difficult to estimate

The Two-way Cross-classified Model with Interaction

- Any level-1 unit (i) is necessarily nested in one specific combination of k and j , while the combinations (kj) are cross-classified in clusters k and j

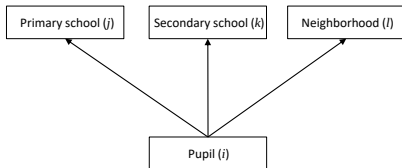
A two-way cross-classified model with interaction between REs:



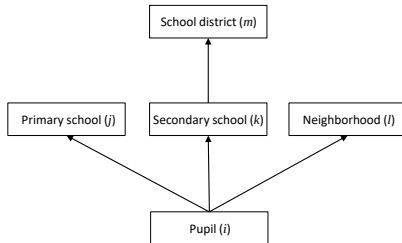
More Complex Structures

- Cross-classified models can have multiple clusters and additional hierarchical nesting in higher-level units:

Three-way cross-classified model:



Three-way cross-classified model with additional strict nesting:



R Example: Student Attainment

- We are using data from the Scotland Neighbourhood Study (Garner and Raudenbush 1991)
 - 524 neighborhoods, 17 secondary schools, 2,310 students
 - dependent variable: attainment score at end of compulsory schooling (age 16)
 - independent variable: Neighborhood deprivation - a neighborhood-level variable measuring the concentration of poverty, bad housing and bad health status
 - controls: attainment measures at age 12, sex, socio-economic background
 - all continuous variables are standardized
 - Example from Leckie (2012)

R Example: Between School Differences in Attainment

```
# Two-level model with strict nesting of students in schools
model2l0 <- lmer(formula=attain ~ 1 + (1|schid), data=sns)
# Cross-classified model with school and neighborhood REs
modelcc0 <- lmer(formula=attain ~ 1 + (1|schid) + (1|neighid), data=sns)
# Adding a random effect for interactions between schools and neighborhoods
modelccia0 <- lmer(formula=attain ~ 1 + (1|schid) + (1|neighid) + (1|schid:neighid), data=sns)
```

Table 1: Example: Student attainment with SNS Data - empty models

	2-level model	Cross-classified model	Cross-classified with interaction
(Intercept)	-0.01 (0.08)	-0.02 (0.07)	-0.02 (0.07)
AIC	6441.81	6358.45	6356.31
BIC	6459.04	6381.43	6385.03
Log Likelihood	-3217.90	-3175.22	-3173.15
Num. obs.	2310	2310	2310
Num. groups: schid	17	17	17
Var: schid (Intercept)	0.10	0.08	0.08
Var: Residual	0.93	0.80	0.78
Num. groups: neighid		524	524
Var: neighid (Intercept)		0.14	0.09
Num. groups: schid:neighid			784
Var: schid:neighid (Intercept)			0.07

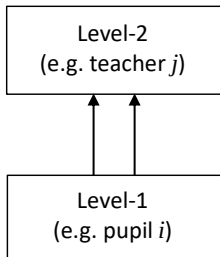
*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

R Example: Interpretation

- When we do not account for neighborhood effects, school-level effects are overestimated
- If we trust the BIC, the model with additional random interaction effects is not better than the model without them
- For our subsequent analysis we will, thus, use a two-way cross-classified model. Note, however, that ignoring RE interactions can bias the estimation of the remaining parameters (Shi, Leite, and Algina 2010)
- We will use this data in Exercise 7, but before we briefly talk about multiple membership models

Multiple Membership Models

- Remember: Multiple membership (MM) models are similar to cross-classified models
 - no strict hierarchical nesting: level-1 units are nested in multiple level-2 units
 - in MM models, units are nested in multiple higher-level clusters of the same type
 - example: Pupils visit classes from different teachers



The Multiple Membership Model

The multiple membership model:

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + \sum_{j=1}^J w_{j,i} \cdot u_j + e_{ij}$$

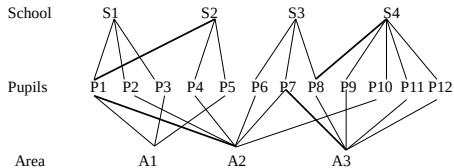
- $\sum_{j=1}^J w_{j,i} \cdot u_j$ is a weighted sum of school effects, where the weight $w_{j,i}$ measure to what extend unit i belongs to the higher-level cluster j
 - $w_{j,i}$ is defined so that is sums to one for each unit i :

$$\sum_{j=1}^J w_{j,i} = 1 \text{ for each } i$$

- This is, $w_{j,i}$ measures what proportion of time i spends in the different clusters j

Remember Our Example of Students in Schools

A unit-diagram of pupils nested in schools and neighborhoods



- We ignore neighborhoods
- And focus only on students P1, P2, P7 and P8

Weights in Multiple Membership Data

A data matrix measuring $w_{j,i}$

P_i	S1	S2	S3	S4
1	0.8	0.2	0.0	0.0
2	1.0	0.0	0.0	0.0
7	0.0	0.0	1.0	0.0
8	0.0	0.0	0.5	0.5

- Pupil P1 switched between schools S1 and S2
 - S/he spend 80% of time in school S1 and 20% in S2
- Pupils P2 and P7 did not switch
 - P2 spend 100% of time in school S1
 - P7 spend 100% of time in school S3
- Pupil P8 switched between S3 and S4
 - 50% time spend in S3 and 50% in S4

How To Estimate Such Models in R

- Unfortunately, MM models cannot be estimated 'out of the box' by lme4
- The **lmerMultiMember**-package provides a wrapper around the lmer-function
 - it adds an argument of the type
`memberships=list(cluster=membership_matrix)`, where
cluster is the name of the cluster-level variable (e.g. schoolID)
- We leave it here and do not consider these models any further

Exercise 7: Analyzing School and Neighborhood Effects in R

- Open the PDF “Exercise 7”; it will walk you through an analysis of the SNS data.

Generalized Mixed Models

- So far we have only talked about linear models
 - such models require a continuous dependent variable
 - many variables of interest, however, are categorical
- Examples:
 - having a university degree (yes/no)
 - being unemployed (yes/no)
 - voted in the last election? (yes/no)
 - which party voted for? (Party A, Party B, Part C)
 - social class (lower, middle, upper)
 - having children (yes/no)
 - how much immigration should government allow? (much, some, none)
 - ...
- We are now talking about models for nominal and ordinal response variables

How To Model Categorical Data

- With categorical data, standard linear regression runs into problems
- Consider the dichotomous variable 'voted in the last election' (0=no, 1=yes)
 - if we treat this as a dependent variable, we essentially want to model a probability ($Pr(y_1 = 1) = \pi_i$), i.e. the probability of having voted
 - a linear regression model would look like this:
$$y_i = \beta_0 + \beta_1 x_{1i} + e_i$$
 - however, it can be shown that the assumptions about the error term ($e_i \sim N(0, \sigma^2)$ and $Var(e_i | x_{1i}, \dots, x_{ki}) = \sigma^2$) are violated with data that follow a Bernoulli distribution
 - moreover, the estimated function $\beta_0 + \beta_1 x_{1i} + e_i$ can take any value between $-\infty$ and $+\infty$, but a probability, obviously, is limited to a range of values ($\pi_i \in [0, 1]$)

The Generalized Linear Model

A solution to this problem is to apply a transformation to the model:

$$\pi_i = F(\beta_0 + \beta_1 x_{1i})$$

- F is a transformation function, which maps the probability π to any value in the range $[-\infty, +\infty]$
- F is (usually) the cumulative distribution function of a distribution with known mathematical properties
 - logistic distribution ($\rightarrow$ logistic regression)
 - standard normal distribution ($\rightarrow$ probit regression)
- The generalized linear model (GLM) uses the inverse of F (F^{-1}) to derive a model that is linear in its parameters (β_0, β_1) :

$$F^{-1}(\pi_i) = \beta_0 + \beta_1 x_{1i}$$

Link Functions in GLMs

- F^{-1} is called the link function because it links π to the regression function
 - the easiest link function is the identity function, where $F^{-1}(\pi) = \pi$. The identity link function leads us back to the linear probability model
 - the logistic function:

$$\pi_i = F(\beta_0 + \beta_1 x_{1i}) = \frac{\exp(\beta_0 + \beta_1 x_{1i})}{1 + \exp(\beta_0 + \beta_1 x_{1i})}$$

- leads to the logit link

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 x_{1i}$$

- other link functions are
 - the inverse of the cumulative distribution function of the standard normal distribution (ϕ^{-1}), which leads to the probit model
 - the inverse of the cumulative distribution function of a Gumbel distribution, which leads to the complementary log-log model
 - in the social sciences, the logit model is the most prominent
 - we consider only the logit model in the remainder of this session

How To Interpret the Logit Model

Coefficients in the logit model give the change in the log-odds:

$$\log \left(\frac{\pi_i}{1 - \pi_i} \right) = \beta_0 + \beta_1 x_{1i}$$

Applying the exponential functions to both sides of the equation gives this:

$$\frac{\pi}{1 - \pi} = \exp(\beta_0 + \beta_1 x_{1i}) = \exp(\beta_0) \cdot \exp(\beta_1 x_{1i})$$

- We call these coefficients **odds ratios**, they tell us by what factor the odds $\left(\frac{\pi_i}{1 - \pi_i} \right)$ change

- GLMs are typically estimated with maximum likelihood
- Other algorithms are available
- But there is no analytical solution to them, as with standard OLS regression
- Parameters can only be estimated in an iterative process of numerical optimization

Introducing the Two-level Logistic Model with RI

Our logit model can be expanded to include a random intercept at level-2:

$$\log \left(\frac{\pi_{ij}}{1 - \pi_{ij}} \right) = \beta_0 + \beta_1 x_{1ij} + u_j$$

- Since there is no idiosyncratic error in these non-linear models, we only have to make an assumption about u :

$$u_j \sim N(0, \sigma_{u0}^2)$$

- Interpretation of σ_u^2 is identical to the linear model, but now it is variation in the log odds

Judging the Amount of Cluster-level Variation from Null Model

- With non-linear models, it is not straightforward to calculate the VPC, because the error term at the individual level is fixed (in the logit model it is always $\frac{\pi^2}{3}$)
 - One alternative approach to judge the amount of variation between clusters is to calculate the predicted probabilities for different values of u_j :

$$\pi_{ij} = \frac{\exp(\beta_0 + u_j)}{1 + \exp(\beta_0 + u_j)}$$

- for example we might take the values $+\sigma_u$ and $-\sigma_u$ (or $2 \cdot \sigma_u$ and $-2 \cdot \sigma_u$), to get predicted probabilities for clusters one (two) standard deviation(s) below and above the overall intercept
- the VPC ($\frac{\sigma_u^2}{\sigma_u^2 + \frac{\pi^2}{3}}$) has only meaning in the null model

The Scaling Issue

- This is only a very brief explanation. To really understand this, it is helpful to consider the so called latent variable formulation of the logistic model (Agresti 2012)
- Anyway, the idiosyncratic error in non-linear models is fixed (e.g. to 1 in the probit model and to $\frac{\pi^2}{3}$ in the logit model)
 - thus, adding variables to your model, cannot reduce (unexplained) level-1 variance
 - imagine we add a level-1 variable (x_{1ij}) that explains variation only within clusters j . In the linear model, this would result in a reduction of level-1 variance (σ_e^2) and **no** change in level-2 variance (σ_u^2)
 - in non-linear models, the residual variance at level-1 is fixed, but since we explained some of it, the remaining coefficients are rescaled, which will make the estimate of the level-2 variance (σ_u^2) larger

- Rescaling also influences coefficients, meaning that we cannot compare coefficients between models (Allison 1999)
 - An accessible discussion of the problem is done by Mood (2010)
 - A popular solution has been proposed by Karlson, Holm, and Breen (2012), also see Breen, Karlson, and Holm (2021) (also implemented in R with the khb package)
- Important for us: We cannot easily compare REs (i.e. variance components) and FEs (i.e. coefficients) across models
 - it is not possible to simply quantify the amount of explained variation at the different levels
 - we do not go into the details of the proposed solutions but simply refrain from interpreting variance components

Let's Use R - the glmer Function

- Next to lmer, lme4 has a second important function: **glmer**
 - glmer estimates generalized linear mixed (regression) models and is equivalent to lmer in many respects, but ...
 - glmer adds the argument `family`, which specifies the link function
- Examples:
 - `family=binomial(link="logit")`, specifies a logistic model
 - `family=binomial(link="probit")`, specifies a probit model
 - `family=gaussian(link="identity")`, specifies a linear model
 - `family=poisson(link="log")`, specifies a poisson model (for count data with a poisson distribution)
 - and many more

R Example: Anti-immigrant Attitudes in Europe

- We are working with data you already know: ESS data on anti-immigrant attitudes
- For this demonstration of the glmer function, I simply dichotomized the dependent variable

R Example: Estimating an Empty Two-level Logistic Model with R

```
# Empty model of anti-immigrant attitudes
model0 <- glmer(formula=ais ~ 1 + (1|Country), data=esslogit, family=binomial(link="logit"))
summary(model0)

## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula: ais ~ 1 + (1 | Country)
## Data: esslogit
##
##          AIC      BIC   logLik deviance df.resid
## 11951.2 11965.5 -5973.6 11947.2     9352
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.9643 -0.7876 -0.6449  0.8665  2.4204
##
## Random effects:
##  Groups Name      Variance Std.Dev.
##  Country (Intercept) 0.6251  0.7907
## Number of obs: 9354, groups: Country, 21
##
## Fixed effects:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.1297    0.1744  -0.744   0.457
```

R Example: How Much Variation Between Countries?

Remember that the residual variance at level-1 is $\frac{\pi^2}{3}$; the VPC is thus:

```
# Calculate VPC  
.6251/(.6251+(pi^2/3))
```

```
## [1] 0.1596692
```

Calculating the probability to be anti-immigrant in countries one standard deviation below and one standard deviation above average:

```
# Overall probability to be anti-immigrant  
exp(-.1297)/(1+exp(-.1297))
```

```
## [1] 0.4676204
```

```
# Probability to be anti-immigrant in a country one standard deviation above average  
exp(-.1297+.7907)/(1+exp(-.1297+.7907))
```

```
## [1] 0.659485
```

```
# Probability to be anti-immigrant in a country one standard deviation below average  
exp(-.1297+-.7907)/(1+exp(-.1297+-.7907))
```

```
## [1] 0.2848764
```

R Example: Adding Variables

```
# Empty model of anti-immigrant attitudes
modell1 <- glmer(formula=ais ~ 1 + res_mig + res_educ + lrscale + gndr
                + agea + (1|Country), data=esslogit, family=binomial(link="logit"))
modell2 <- glmer(formula=ais ~ 1 + res_mig + res_educ + lrscale + gndr
                + agea + infl_for_perc_inter + (1|Country), data=esslogit,
                family=binomial(link="logit"))
```

Table 3: Example: Logistic Multilevel Model of Anti-Immigrant Attitudes

	M0	M1	M2
Intercept	-0.13 (0.17)	-0.82 (0.17)***	-0.64 (0.15)***
Migration Background (Ref.=no)		-0.29 (0.06)***	-0.29 (0.06)***
Upper Medium Educ. (Ref.=high)		0.48 (0.08)***	0.48 (0.08)***
Lower Medium Educ. (Ref.=high)		0.98 (0.06)***	0.98 (0.06)***
Low Educ. (Ref.=high)		1.10 (0.07)***	1.10 (0.07)***
Left-Right		0.31 (0.02)***	0.31 (0.02)***
Female (Ref.=Men)		0.07 (0.05)	0.07 (0.05)
Age		0.12 (0.02)***	0.12 (0.02)***
Inflow Immigrants			-0.36 (0.10)***
AIC	11951.23	11320.53	11313.02
BIC	11965.52	11384.82	11384.45
Log Likelihood	-5973.62	-5651.26	-5646.51
Num. obs.	9354	9354	9354
Num. groups: Country	21	21	21
Var: Country (Intercept)	0.63	0.57	0.34

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Summary: Generalized Linear Mixed Models

- Generalized linear models is an extensive topic, which can easily fill multiple lectures
 - we just scratched the basic idea, but there are other courses
 - Faraway (2016) treats GLMs and mixed models in R
 - interpretation in non-linear models is generally more complicated
- The single-level GLM can easily be extended with random intercepts and random slopes
 - much of what we learned about the linear model applies here
 - interpretation of variance components, however, is difficult
- Many different link functions are available
 - ordinal logit/probit model not implemented in lme4 yet, but there are alternatives (e.g. the ordinal package)
 - multinomial logit model also not in lme4, but for example in mcllogit package

Exercise 8: Estimating GLMs Models in R

Not today, tomorrow: Open the PDF “Exercise 8”; it will walk you through one analysis task.

Session 5: Analyzing panel data (Part I)

Today:

- (Exercise 8: Generalized linear multilevel models in R)
- Introduction to panel data
- Omitted variable bias
- Wide and long format
- The fixed effects model
 - the LSDV model
 - the FE transformation
- Exercise 9: Panel data management and FE models in R

Panel Data - What's That And What Can We Do With It?

Panel data contains **repeated observations** of the same units ($i = 1, \dots, n$) over time ($t = 1, \dots, T$). Such data comes with many opportunities, but also with statistical challenges

What can we do with panel data?

- Describe and explain developments over time, for example ...
 - how does social spending of nations develop over time?
 - how does life satisfaction after a divorce develop?
 - how do earning trajectories of men and women over their career look like?
- Get better estimates of causal effects, for example ...
 - does immigration have an effect on social spending?
 - does unemployment decrease individuals' life satisfaction?
 - does union membership increase individual wages?

Panel Data - How Does It Look?

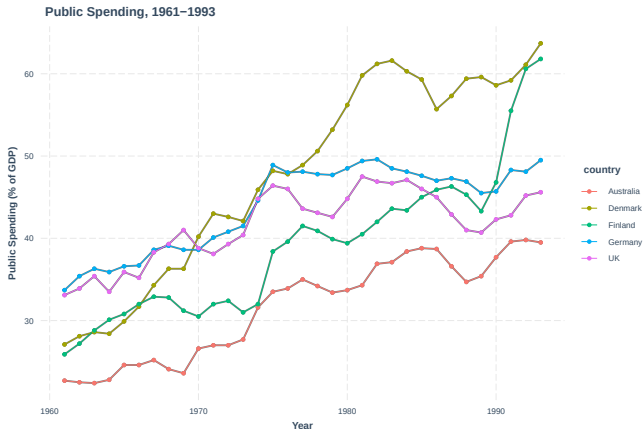
Panel data looks like this:

Table 4: Artificial panel data in long format

Country	Year	y	x1	x2
Germany	2010	6.57	4.55	3.72
Germany	2011	4.82	8.58	6.50
Germany	2012	5.01	7.25	5.60
Germany	2013	4.45	9.96	6.18
France	2010	4.55	5.53	4.65
France	2011	3.03	3.56	7.93
France	2012	4.57	3.75	4.00
France	2013	2.54	7.90	2.12

Panel Data - How Does It Look?

Panel data provides time-series for multiple units:



Panel Data - Just Clustered Observations?

- Previously, we have learned that multilevel models account for clustered observations
- For example:
 - students in classes and/or schools
 - individuals in countries
 - patients in hospitals
 - repeated observations of individuals
- Is there anything special about panel data, then? Yes, ...
 - ... with panel data the units of analysis (i) are observed at level-2
 - ... at level-1, the **order** of the observations t is **important**

Some Terminology

The term **panel data** is most often used for survey data on individuals. An alternative name for panel data is **Pooled Time Series Cross Section data** (PTSCS), typically used when units of analysis are countries.

- Panel data: Typically has large sample size (n), with relatively few repeated observations (T).
 - often a (large) random sample of individuals
 - followed over a couple of years ($n > T$)
- PTSCS data: Usually has much smaller n , but longer observations periods.
 - typically not a random sample but a selection (e.g. OECD countries, European countries, etc.)
 - often $n < T$
- Other terms: Longitudinal data, repeated measures data

Again Let's Start With a Reminder of the OLS Model

What does standard ordinary least square (OLS) regression do?

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + e_i$$

The 'magic' happens with the error term. We estimate the parameters $(\beta_0, \beta_1, \beta_2)$ that minimize the difference between the predicted value of y :

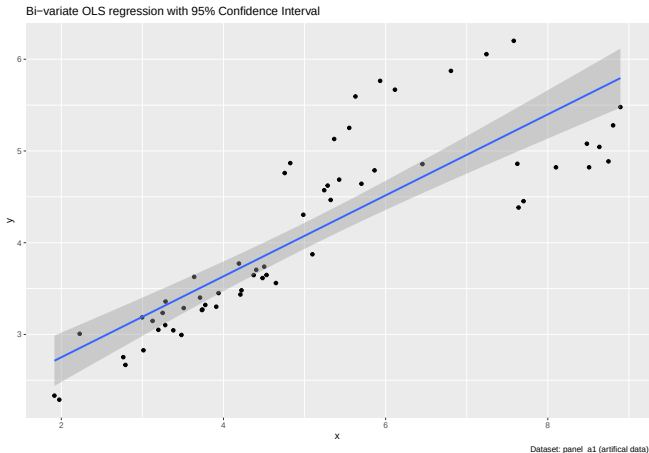
$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i}$$

and the individual observations y_i : $y_i - \hat{y}_i$, which is to minimize the sum of the squared residuals:

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n e_i^2$$

An OLS model in a Bivariate Graph

A simple OLS regression with some data, where the slope of the regression line corresponds to β :



OLS Assumptions: When Does It Work?

For an OLS to work, a number of assumptions have to be met (Andreß, Golsch, and Schmidt 2013, 122f.):

1. Data are a simple random sample.
2. The (true) model is linear in its parameters.
3. Each x-variable is neither constant nor a linear combination of other x-variables.
4. The error term is independent of the variables in the model:
$$E(e_i | x_{1i}, \dots, x_{ki}) = 0.$$
5. The error is homoscedastic, i.e. has a constant variance:
$$\text{Var}(e_i | x_{1i}, \dots, x_{ki}) = \sigma^2.$$
6. The errors are independent of each other, i.e. they do not correlate between any two units i and j ($i \neq j$):
$$\text{Corr}(e_i, e_j | x_{1i}, \dots, x_{ki}) = 0.$$
7. The error term has a normal distribution: $e_i \sim N(0, \sigma^2)$.

Isn't that the same story as in session 2?

- In parts yes!
- But with panel data there is a different focus!

The Exogeneity Assumption

For an OLS to work, a number of assumptions have to be met (Andreß, Golsch, and Schmidt 2013, 122f.):

1. Data are a simple random sample
2. The (true) model is linear in its parameters
3. Each x-variable is neither constant nor a linear combination of other x-variables
4. The error term is independent of the variables in the model:
 $E(e_i | x_{1i}, \dots, x_{ki}) = 0.$
5. The error is homoscedastic, i.e. has a constant variance:
 $Var(e_i | x_{1i}, \dots, x_{ki}) = \sigma^2$
6. The errors are independent of each other, i.e. they do not correlate between any two units i and j ($i \neq j$):
 $Corr(e_i, e_j | x_{1i}, \dots, x_{ki}) = 0$
7. The error term has a normal distribution: $e_i \sim N(0, \sigma^2)$

OLS - the Best Linear Unbiased Estimator

The Gauss-Markov-Theorem tells us: If these assumptions are met, OLS is BLUE - the **B**est **L**inear **U**nbiased **E**stimator (Wooldridge 2015).

- **Best** refers to the standard errors:
 - OLS is the most efficient estimator
 - this is, it has the smallest standard errors/ confidence intervals
- **Unbiased** refers to the point estimates, i.e. coefficients:
 - OLS will provide the true population parameter
 - this is, the estimated effects ($\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$) equal the true effects
 - not necessarily in a single sample, but on average

OLS Assumptions and Statistical Inference

- Some of the assumptions are necessary to draw correct statistical inference from a model. Assuming the model is correctly specified, assumptions 1, 5, 6, and 7 are necessary to get correct standard errors
 - random sample: Often not the case with PTSCS data, or, more generally, with cross-national data (Schmidt-Catran, Fairbrother, and Andreß 2019), but we can get around that by citing Beck (2001): ‘all inferences of interest are conditional on the observed units’
 - homoscedastic and normally distributed error term is necessary, but this can be tested and/or modeled
 - with panel data, error terms are not independent and we already learned about that

OLS Assumptions and Bias in Coefficients

- Assumption 4 ($E(e_i|x_{1i}, \dots, x_{ki}) = 0$) is necessary to get unbiased coefficients. It is often called the **exogeneity assumption** and it is **very important**:
 - the variables in our model ($x_1, x_2, \dots, x_k$) must not correlate with the error term
 - basically, we must specify a correct model and cannot miss any important variables that are related to the variables of interest, or else we will get biased estimates. We call this **omitted variable bias**
 - so far, we have not talked about bias at all; with panel data, this will be our focus
- Panel data can help us to avoid omitted variable bias. We'll see that later! First, let's understand omitted variable bias. . .

Let's Create a Population Model

- Assume we have a population in which the following model is correctly specified:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + e_i$$

- The true parameters are $\beta_0 = 2$ and $\beta_1 = 1$ and $\beta_2 = 1$:

$$y_i = 2 + 1x_{1i} + 1x_{2i} + e_i$$

- Furthermore, we assume that x_1 and x_2 are correlated:

$$\text{Corr}(x_1, x_2) = .5$$

- What happens when we ignore x_2 in our model specification?
Let's see. . .

Let's Create Some Data

```
set.seed(1)
# create variance-covariance matrix and mean vector of x1 and x2
sigma <- rbind(c(1, 0.5), c(0.5, 1))
mu <- c(5, 5)
# generate the multivariate normal distribution of x1 and x2
df <- as.data.frame(mvrnorm(n=1000, mu=mu, Sigma=sigma))
colnames(df) <- c("x1", "x2")
# Generate y
df$y <- 2 + 1*df$x1 + 1*df$x2 + rnorm(1000, 0, .5)
head(df, n=4L)
```

```
##           x1           x2           y
## 1 3.889993 5.024958 10.47188
## 2 4.603074 5.715006 11.35695
## 3 4.711713 3.840936 11.36250
## 4 6.276188 6.486919 15.02274
```

```
cor(df$x1, df$x2)
```

```
## [1] 0.4963366
```


Let's Estimate Some OLS Models

```
ols_true      <- lm(y ~ x1 + x2, data=df)
ols_mis_spec  <- lm(y ~ x1, data=df)
screenreg(list(ols_true, ols_mis_spec))
```

```
##
## =====
##              Model 1          Model 2
## -----
## (Intercept)      1.87 ***      4.42 ***
##                  (0.09)        (0.17)
## x1                1.00 ***      1.51 ***
##                  (0.02)        (0.03)
## x2                1.03 ***
##                  (0.02)
## -----
## R^2              0.93          0.69
## Adj. R^2         0.93          0.69
## Num. obs.        1000          1000
## =====
## *** p < 0.001; ** p < 0.01; * p < 0.05
```

Omitted Variable Bias

- Model 2 is misspecified, falsely attributing some of x_2 's effect to x_1 . Since $\text{Corr}(x_1, x_2) = .5$ we get

$$\hat{\beta}_1 = \beta_1 + .5 \cdot \beta_2 = 1 + .5 \cdot 1 = 1.5).$$

```
texreg(list(ols_true, ols_mis_spec), no.margin=TRUE, caption.above=TRUE,  
caption="Omitted variable bias")
```

Table 5: Omitted variable bias

	Model 1	Model 2
(Intercept)	1.87*** (0.09)	4.42*** (0.17)
x1	1.00*** (0.02)	1.51*** (0.03)
x2	1.03*** (0.02)	
R ²	0.93	0.69
Adj. R ²	0.93	0.69
Num. obs.	1000	1000

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Omitted Variables and the Exogeneity Assumption

- How does the error term of Model 2 look like?
 - the population model was:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + e_i$$

- if we estimated $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i}$, where did x_2 go?
 - into the error term of our estimated model:

$$\hat{e}_i = e_i + \beta_2 x_{2i}$$

- since $\text{Corr}(x_1, x_2) \neq 0$, it follows that the error term is correlated with the variables in the model ($E(\hat{e}_i | x_{1i}) \neq 0$)

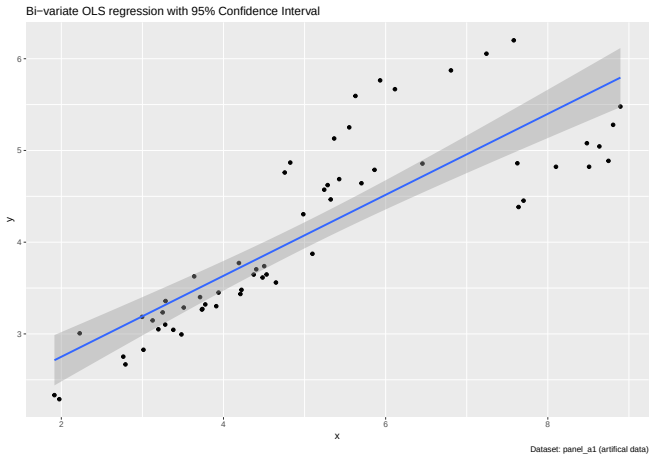
Summary: The OLS model

- What we want:
 - unbiased coefficients, aka point estimates
 - standard errors that are efficient but not anti-conservative
 - this is, BLUE
- What we need:
 - a correctly specified model with no omitted variables
 - independent observations
 - A bunch of other things (like homoscedastic and normally distributed error terms)
- What we learned before:
 - how to correct for non-independent observations via random intercepts
 - how to model heteroscedasticity via random slopes

Now, let's look at panel data to learn how we can correct for omitted variable bias

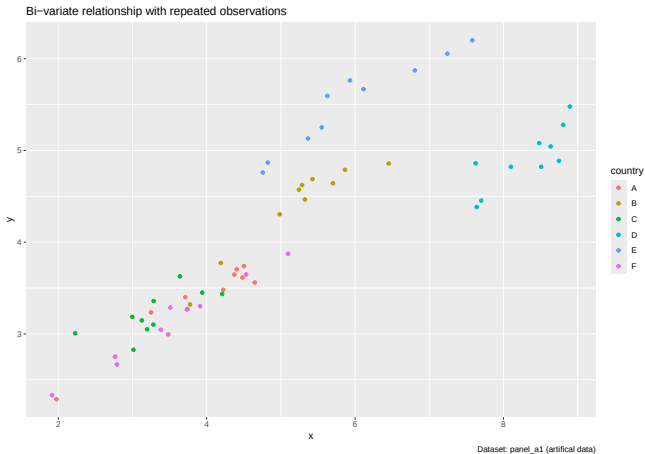
The OLS Model Again

Here is our graph of the bi-variate OLS regression again:



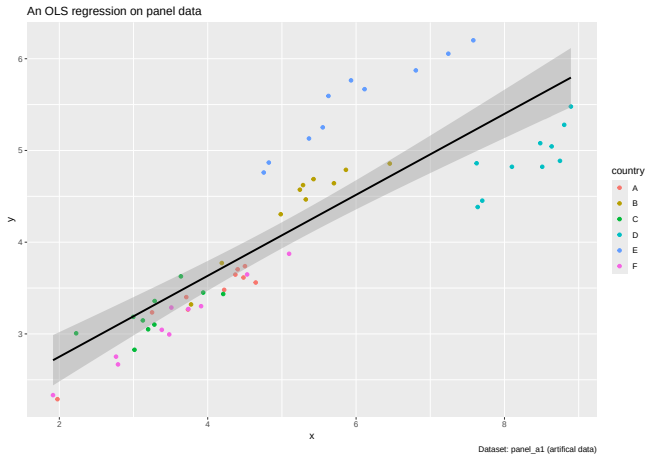
Panel Data in a Bi-variate Graph

But what, if the data is actually panel data?



OLS Regression on Panel Data

Are these error terms independent?



OLS Provides Downward Biased SEs With Panel Data

- Just as before, we get underestimated standard errors from clustered data:

```
ols      <- lm(y ~ x, data=panel_a1)
ols_crse <- lm.cluster(y ~ x, data=panel_a1, cluster = 'country')
```

Table 6: OLS on panel data with normal and robust standard errors

	Model 1	Model 2
(Intercept)	1.87*** (0.20)	1.87*** (0.47)
x	0.44*** (0.04)	0.44*** (0.12)
R ²	0.70	0.70
Adj. R ²	0.70	0.70
Num. obs.	60	60

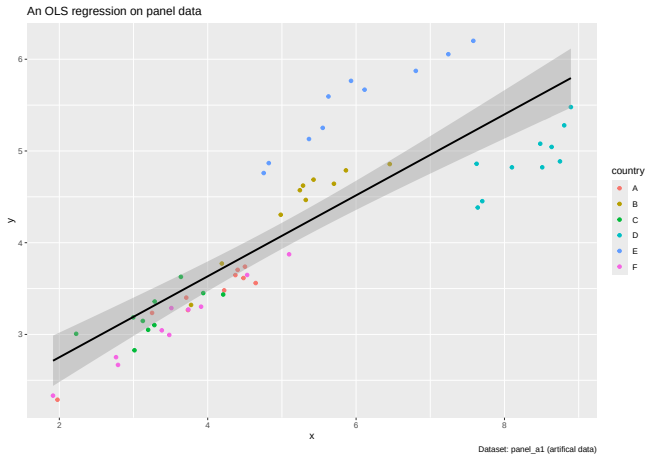
*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Bias in Standard Errors

- With panel data, the dependency in the observations will often be much stronger than with other kinds of multilevel data
- Examples of multilevel data:
 - students in the same class share some similarities, but still there will be a lot of differences between students
 - individuals in the same countries will share some similarities, but there will be still a lot of differences between individuals
 - typical ICCs lie between .05 and .2
- Examples of panel data:
 - repeated observations of the same individuals' employment status, will have the same value for most individuals
 - repeated observations of the same countries unemployment rates will have very high correlations
 - typical ICCs depend strongly on the dependent variable, but can easily exceed .5

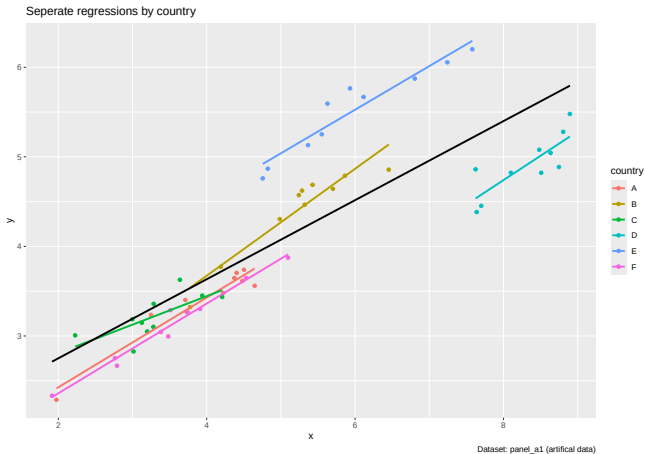
OLS Regressions on Panel Data

What about bias? Does the OLS regression provide a good estimate of the effect of x on y ?



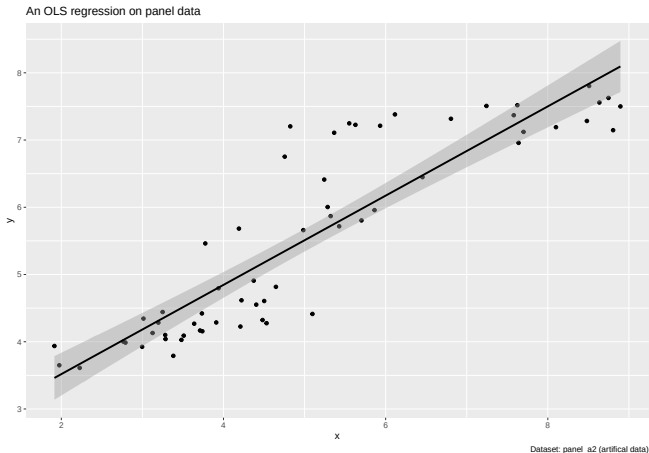
OLS Regressions by Country

Maybe bias is not a problem, because every possible regression tells the same story.



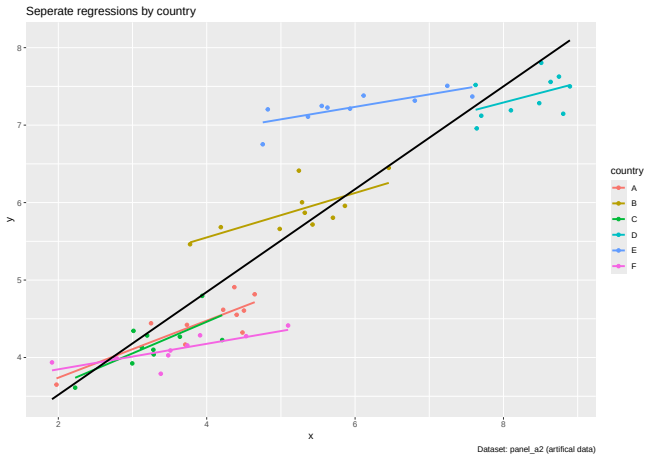
OLS on Another Example Data Set

What about this one? Does OLS provide a good estimate of the effect of x on y ?



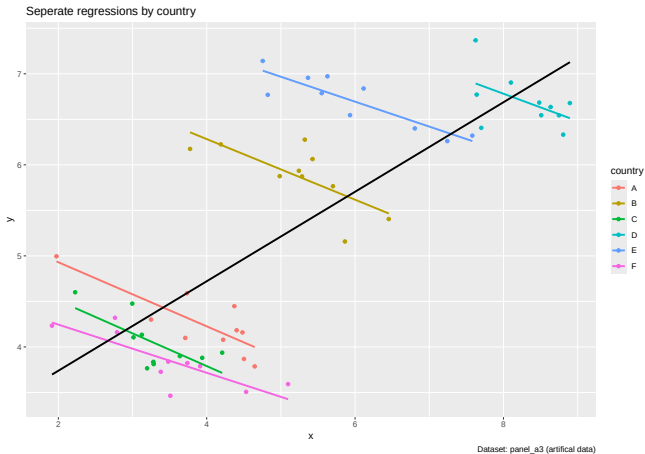
OLS Regressions by Country

Maybe not ...



OLS Regressions by Country

And this one? OLS can be severely biased!



Bias in Observational Studies

- Causal effects of x on y are 'easily' identified with experimental studies
- Observational data makes it much harder to identify causal effects
- Cross-sectional regression type models ...
 - identify effects from comparisons between units of analysis
 - have a high risk of omitted variable bias
- Panel data allow ...
 - to make use of variation within units of analysis
 - to avoid omitted variable bias (if it's due to unobserved unit-heterogeneity) and get better estimates of causal effects

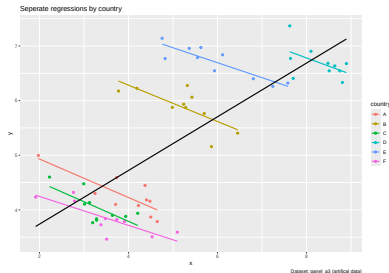
Our First Fixed Effects Panel Model

```
ols <- lm(y ~ x, data=panel_a3)
fixed <- plm(y ~ x, data=panel_a3, index=c("country", "year"), model="within")
```

Table 7: OLS vs. FE model

	OLS	Fixed Effects
(Intercept)	2.76*** (0.32)	
x	0.49*** (0.06)	-0.31*** (0.04)
R ²	0.55	0.55
Adj. R ²	0.54	0.50
Num. obs.	60	60

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$



Panel Data - Wide and Long Format

- Panel data can be seen as a three-dimensional data cube, where n units i are observed at T time points t
- There are two ways to integrate panel data into a two-dimensional data matrix
 - in the **wide** format time is integrated into the columns/variables of the data matrix
 - in the **long** format time is integrated into the rows of the data matrix

Panel data in long Format

- In long format, panel data needs an unit-ID and an additional identifier for time
- Data set has $n \cdot T$ rows and V columns, where V is the number of variables

Table 8: Artifical panel data in long format

Country	Year	y	x1
Germany	2012	111.1	14.0
Germany	2013	105.8	39.3
France	2012	119.4	48.1
France	2013	87.9	29.4

Panel Data in Wide Format

- In wide format, time **must** be integrated into the variable names
- The data set has $T \cdot V$ columns and n rows

Table 9: Artificial panel data in wide format

Country	y_2012	x1_2012	y_2013	x1_2013
France	119.4	48.1	87.9	29.4
Germany	111.1	14.0	105.8	39.3

When to use which format?

- For (most) regression type analysis, data needs to be in long format
 - this is also what we have been using so far, but we never talked about it
- For descriptive purposes and some other techniques (e.g. structural equation modeling), data in wide format is better suited/ required
 - for example, it is much easier to calculate the change from t to $t + 1$ in wide format than it is in long format

Going From Long to Wide Format

Panelr (Long 2020) is a very helpful package. We will use it intensively later. For now, we have a look at its data management commands:

```
# Display panel data set in long format  
panel_long
```

```
##   Country Year      y      x1  
## 1  France 2012 119.44596 48.06089  
## 2  France 2013  87.94662 29.43496  
## 3 Germany 2012 111.08672 13.97374  
## 4 Germany 2013 105.79913 39.33251
```

Going from Long to Wide Format (continued)

```
# Declaring the data frame 'panel_long' to be a panel data object
panel_long <- panel_data(panel_long, id=Country, wave=Year)
# Convert to wide format and display
panel_wide <- widen_panel(panel_long)
panel_wide
```

```
## # A tibble: 2 x 5
##   Country y_2012 x1_2012 y_2013 x1_2013
##   <fct>    <dbl>   <dbl>   <dbl>   <dbl>
## 1 France    119.    48.1    87.9    29.4
## 2 Germany   111.    14.0   106.    39.3
```

Going from Wide to Long Format

- Reshaping from wide to long is more complex, as there are many ways to integrate time into the variable names
- **panelr** needs to know how time is indicated

```
# Convert (back) to long format and display
panel_long <- long_panel(panel_wide, prefix="_", begin=2012, end=2013,
                          label_location="end")

panel_long
```

```
## # Panel data:      4 x 5
## # entities:      id [2]
## # wave variable: wave [2012, 2013 (2 waves)]
##   id    wave Country    y    x1
##   <fct> <dbl> <fct>    <dbl> <dbl>
## 1 1      2012 France  119.   48.1
## 2 1      2013 France   87.9  29.4
## 3 2      2012 Germany 111.   14.0
## 4 2      2013 Germany 106.   39.3
```

Data Mangement in panelr - Summary

Important commands from the panelr package:

- `panel_data()` declares data frames to be panel data objects
- `widen_panel()` reshapes from long to wide format
- `long_panel()` reshapes from wide to long format

From Simple OLS to a Panel Model

- Once again, remember the OLS model, which assumes independent observations (aka a simple random sample):

$$y_i = \beta_0 + \beta_1 x_{1i} + e_i$$

- With panel data, we have repeated observations of the same units i . Let's add an index t :

$$y_{it} = \beta_0 + \beta_1 x_{1it} + e_{it}$$

- Furthermore, we can introduce the differentiation of time-constant and time-varying variables:

$$y_{it} = \beta_0 + \beta_1 x_{1it} + \gamma_1 z_{1i} + e_{it}$$

- note the slight difference in notation (it instead of ij), which uses not only different letters, but also a different order (from higher to lower level: it instead of ti)

From Simple OLS to a Panel Model

What is an error term?

- e_i is often called the **idiosyncratic** error
- It captures any unobserved effect on y
- It is assumed to be random (aka independent of the variables in the model)
- With panel data, we can differentiate between time-constant and time-varying variables and we can do the same with the error term:

$$y_{it} = \beta_0 + \beta_1 x_{1it} + \gamma_1 z_{1i} + u_i + e_{it}$$

The Unobserved Effects Model

In the language of 'panel people', this model is called an unobserved effects model:

$$y_{it} = \beta_0 + \beta_1 x_{1it} + \gamma_1 z_{1i} + u_i + e_{it}$$

- it assumes that the dependencies between observations from the same unit are due to unit-specific characteristics
- u_i is often called **unobserved heterogeneity**
- if u_i is correlated with the variables in the model, estimates will be biased

How to Get Rid of Unobserved Heterogeneity?

Unobserved heterogeneity can be a source of serial correlation and biased coefficients. How can we control for it?

- One obvious way is to control for the unit-specific effects u_i via a set of unit-specific dummy variables:

$$y_{it} = \beta_0 + \beta_1 x_{1it} + \sum_{i=1}^{n-1} \delta_i d_i + e_{it}$$

- the unobserved effects are modeled as **Fixed Effects**
- this model controls for any time-constant differences between the units of analysis Estimates are identified from **within-unit variation only**
- any time-constant variables z_i must be dropped because they are perfectly collinear with the set of dummies
- it can be estimated with simple OLS and is often called the **Least Square Dummy Variable** (LSDV) approach to the FE model

Let's Try in R

```
# Estimate OLS model with set of dummies for countries
```

```
lsdv <- lm(y ~ x + factor(country), data=panel_a3)
```

```
summary(lsdv)
```

```
##
## Call:
## lm(formula = y ~ x + factor(country), data = panel_a3)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.52224 -0.12666  0.02477  0.11973  0.47047
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    5.45073    0.16311  33.417  < 2e-16 ***
## x             -0.30532    0.03775  -8.088 8.09e-11 ***
## factor(country)B  2.01964    0.10760  18.770  < 2e-16 ***
## factor(country)C -0.40142    0.09885  -4.061 0.000162 ***
## factor(country)D  3.77470    0.19125  19.737  < 2e-16 ***
## factor(country)E  3.07443    0.12319  24.956  < 2e-16 ***
## factor(country)F -0.53308    0.09715  -5.487 1.17e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2144 on 53 degrees of freedom
## Multiple R-squared:  0.9744, Adjusted R-squared:  0.9715
## F-statistic: 336.6 on 6 and 53 DF,  p-value: < 2.2e-16
```

An Alternative Specification of the Same Model

Instead of partialing out variance between the units of analysis by unit-dummies, we can apply the so called Fixed Effects transformation

- Start with the unobserved effects model:

$$y_{it} = \beta_0 + \beta_1 x_{1it} + \gamma_1 z_{1i} + u_i + e_{it}$$

- Build the between model (averaging for each unit across time):

$$\bar{y}_i = \beta_0 + \beta_1 \bar{x}_{1i} + \gamma_1 \bar{z}_{1i} + \bar{u}_i + \bar{e}_i$$

An Alternative Specification of the Same Model

- Subtract the between model from the unobserved effect model:

$$(y_{it} - \bar{y}_i) = (\beta_0 - \beta_0) + \beta_1(x_{1it} - \bar{x}_{1i}) + \gamma_1(z_{1i} - \bar{z}_{1i}) + (u_i - \bar{u}_i) + (e_{it} - \bar{e}_i)$$

- This reduces to:

$$(y_{it} - \bar{y}_i) = \beta_1(x_{1it} - \bar{x}_{1i}) + (e_{it} - \bar{e}_i)$$

- We call these demeaned variables, but it essentially is group-mean centering
- As in the LSDV approach, ...
 - effects are identified from within-unit variation only
 - time-constant properties (z_1 and u_i) drop out of the equation

Let's Try in R

```
# Do FE transformation by hand
panel_a3_t <- panel_a3 %>% group_by(country) %>%
  mutate(
    y_m = mean(y, na.rm = TRUE),
    x_m = mean(x, na.rm = TRUE)
  ) %>% ungroup() %>%
  mutate(
    y_dm = y - y_m,
    x_dm = x - x_m
  )

# Estimate OLS model with set of dummies for countries
fe_by_hand <- lm(y_dm ~ x_dm - 1, data = panel_a3_t)
summary(fe_by_hand)

##
## Call:
## lm(formula = y_dm ~ x_dm - 1, data = panel_a3_t)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.52224 -0.12666  0.02477  0.11973  0.47047
##
## Coefficients:
##      Estimate Std. Error t value Pr(>|t|)
## x_dm -0.30532    0.03578  -8.534 6.89e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2032 on 59 degrees of freedom
## Multiple R-squared:  0.5524, Adjusted R-squared:  0.5448
```


How to Estimate the Fixed Effects Model?

- Doing the transformation by hand is nice to show that the FE transformation is actually 'under the hood', but it produces **incorrect standard errors**
 - this is because the model does not recognize that it has transformed data
 - the demeaning procedure takes up degrees of freedom and this must be taken into account to get correct inferential statistics
- The **plm** package (Croissant and Millo 2008) has a dedicated command for the FE model
 - in **plm** the FE model is called the within-model. This alternative name refers to the type of variance the model uses for identification
 - the **plm** package also has its own panel data frame object

Let's Try in R

```
# Declaring the data frame 'panel_a3' to be a panel data object
panel_a3_pd <- pdata.frame(panel_a3, index = c("country", "year"), drop.index = TRUE)
# Estimate within-model aka FE model
fixed <- plm(y ~ x, data=panel_a3_pd, model="within")
summary(fixed)
```

```
## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = y ~ x, data = panel_a3_pd, model = "within")
##
## Balanced Panel: n = 6, T = 10, N = 60
##
## Residuals:
##      Min.      1st Qu.      Median      3rd Qu.      Max.
## -0.522240 -0.126663  0.024772  0.119727  0.470473
##
## Coefficients:
##      Estimate Std. Error t-value Pr(>|t|)
## x -0.305322   0.037749  -8.0882 8.093e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    5.4412
## Residual Sum of Squares: 2.4353
## R-Squared:              0.55244
## Adj. R-Squared: 0.50177
## F-statistic: 65.4186 on 1 and 53 DF, p-value: 8.0931e-11
```

Comparing the Three Variants

Comparing the lsdv approach, the FE-by-hand model and the estimates from plm's within model:

Table 10: Three versions of the same FE model

	OLS	FE (by hand)	FE (plm)
(Intercept)	5.4507*** (0.1631)		
x	-0.3053*** (0.0377)		-0.3053*** (0.0377)
factor(country)B	2.0196*** (0.1076)		
factor(country)C	-0.4014*** (0.0988)		
factor(country)D	3.7747*** (0.1913)		
factor(country)E	3.0744*** (0.1232)		
factor(country)F	-0.5331*** (0.0972)		
x_dm		-0.3053*** (0.0358)	
R ²	0.9744	0.5524	0.5524
Adj. R ²	0.9715	0.5448	0.5018
Num. obs.	60	60	60
*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$			

Summary: Panel Data and the FE Model

- Panel data contain repeated observations of the same units
 - assumption of independent observations is violated
 - this leads to correlated residuals
 - panel data models take care of the non-independence of the observations
 - this is what we already knew about multilevel data
- Panel data allows to identify effects from within- instead of between-unit variation
 - this is, the **FE model controls for any time-constant characteristics**
 - it does so by eliminating any between-unit variance, either through a set of dummies for the units of analysis, or by transforming the data
 - the FE model cannot estimate effects of time-constant variables

Exercise 9: Estimating Two-level Models in R

- Open the PDF “Exercise 9”; it will walk you through an analysis task.

Session 6: Analyzing panel data (Part II)

Today:

- A reminder of the FE model
- The Random Effects Model
- How to decide between models?
 - The Breusch-Pagan test
 - The Hausman test
- The within-between and the Mundlak formulation
- Exercise 10: Estimating random effects panel models in R
- Growth curve models

A Brief Reminder of Part I

- Panel data is hierarchically structured data, aka multilevel data
 - it actually can be modeled with the same techniques we learned in sessions 2 and 3 (and we will do that today)
 - the focus of 'panel people', however, is primarily on the estimation of unbiased coefficients
- u_i is called unobserved heterogeneity
 - it contains all unobserved effects of time-constant variables
 - the FE model eliminates u_i from the model equation to control for any of these effects

Let's Look at the Unobserved Effects Model Again

Let's look again at the unobserved effects model:

$$y_{it} = \beta_0 + \beta_1 x_{1it} + \gamma_1 z_{1i} + u_i + e_{it}$$

- Obviously, this looks just like a random intercept (RI) model and we can simply apply what we have learned in sessions 2 and 3...

The Random Effects Model

‘Panel People’ call this the Random Effects (RE) model:

$$y_{it} = \beta_0 + \beta_1 x_{1it} + \gamma_1 z_{1i} + u_i + e_{it}$$

- The model estimates unobserved heterogeneity (u_i), instead of eliminating it from the data
- It assumes that unobserved heterogeneity is **uncorrelated** with the variables in the model
- Time-constant variables z_i can be included

Let's Try in R

We are using the wagepan data set, which covers the period from 1980 to 1987 and contains data on wages (plus some covariates) from 545 individuals.

```
# Declaring the data frame 'wagepan' to be a panel data object (plm)
wagepan_pd <- pdata.frame(wagepan, index = c("nr", "year"),
                           drop.index = TRUE)
# Estimate RE model (ref. for black and hispanic is white)
random <- plm(lwage ~ black + hisp + educ + exper + union,
              data=wagepan_pd, model="random")
# Estimate FE model for comparison
fixed <- plm(lwage ~ black + hisp + educ + exper + union,
             data=wagepan_pd, model="within")
```

The FE model drops all time-constant variables; the RE model can estimate their effects. Thus, the RE model must (obviously) use between-unit variation.

Table 11: RE and FE models of log hourly wages, wagepan data

	RE	FE
(Intercept)	-0.075 (0.111)	
black	-0.158*** (0.048)	
hisp	0.015 (0.043)	
educ	0.111*** (0.009)	
exper	0.063*** (0.002)	0.064*** (0.002)
union	0.113*** (0.018)	0.086*** (0.019)
s_idios	0.354	
s_id	0.328	
R ²	0.165	0.165
Adj. R ²	0.164	0.045
Num. obs.	4360	4360
*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$		

The Breusch-Pagan Test

We can test whether a RE model is necessary, as opposed to a simple OLS model, using the **Breusch-Pagan test**

- Let the total error be $\epsilon_{it} = u_i + e_{it}$
- Using a test by (Breusch and Pagan 1980), we can test $H_0 : \sigma_u^2 = 0$
- This is the same as $H_0 : \text{Corr}(\epsilon_{it}, \epsilon_{is}) = 0$, with $t \neq s$
- And the same as asking whether the ICC is 0
- In the **plm package**, the Breusch-Pagan test is integrated into the command **plmtest**

The Breusch-Pagan Test with the plm Package

```
# Breusch-Pagan test with the wagepan RE model  
plmtest(random, type="bp")
```

```
##  
## Lagrange Multiplier Test - (Breusch-Pagan)  
##  
## data: lwage ~ black + hisp + educ + exper + union  
## chisq = 3230.1, df = 1, p-value < 2.2e-16  
## alternative hypothesis: significant effects
```

- Null hypothesis can be rejected. Thus, there is serial correlation in the data and a RE model must be estimated; OLS would give biased standard errors
- `plmtest` requires the object under which the model is stored (here: `random`) and the type of test (here: `"bp"`)

How to Decide between FE and RE?

- If a panel model is necessary (i.e. Breusch-Pagan test is significant), the next question is whether RE or FE should be estimated
- The **Hausman test** helps to decide between these
 - to put it easy, RE can be estimated if u_i is uncorrelated with the variables in the model, while FE is to be preferred if u_i correlates with the variables in the model
 - to put it more difficult: The Hausman test, compares two estimators which are both consistent under one set of assumptions, but only one of them is consistent under another set of assumptions (Hausman 1978)
 - if RE and FE yield identical results, RE can be estimated. If FE differs from RE, RE is probably biased

The Hausman Test with the plm Package

- The Hausman test has the null-hypothesis that the RE model is to be preferred
- In the **plm package**, the Hausman test is done with the command **phptest**, which needs the two models as arguments

```
# Hausman test comparing th RE and the FE model  
phptest(fixed, random)
```

```
##  
## Hausman Test  
##  
## data: lwage ~ black + hisp + educ + exper + union  
## chisq = 19.853, df = 2, p-value = 4.885e-05  
## alternative hypothesis: one model is inconsistent
```

- Here we need to abandon the RE model and go with the FE model

Why to Prefer RE Model over FE?

- What we know: The RE model allows to include time-constant variables (z_i)
- What we have not discussed:
 - the RE model is more efficient. What does that mean?
 - it has smaller standard errors, i.e. it is more precise
- Why is that? The RE model can be seen as an estimator that lies between the FE and the pooled OLS model:

$$(y_{it} - \theta \bar{y}_i) = (\beta_0 - \theta \beta_0) + \beta_1(x_{1it} - \theta \bar{x}_{1i}) + \gamma_1(z_{1i} - \theta \bar{z}_{1i}) \\ + (u_i - \theta \bar{u}_i) + (e_{it} - \theta \bar{e}_i)$$

, with

$$\theta = 1 - \sqrt{\frac{\sigma_e^2}{\sigma_e^2 + T \cdot \sigma_u^2}}$$

Pooled OLS, RE and FE - How Do They Relate?

The RE model:

$$(y_{it} - \theta \bar{y}_i) = (\beta_0 - \theta \beta_0) + \beta_1(x_{1it} - \theta \bar{x}_{1i}) + \gamma_1(z_{1i} - \theta \bar{z}_{1i}) \\ + (u_i - \theta \bar{u}_i) + (e_{it} - \theta \bar{e}_i)$$

- If $\theta = 1$ this equals the FE model
- If $\theta = 0$ this equals the pooled OLS model

What does that mean?

- The FE model eliminates all between-unit variance
- The RE model uses more variance to estimate the parameters.
The higher the variance, the smaller your standard errors

Pros and Cons of FE and RE

Fixed Effects model

- Pros:
 - identification from within-unit variation
 - controls for unobserved heterogeneity (any constant characteristic)
- Cons:
 - time-constant variables cannot be included

Random Effects model

- Pros:
 - effects of time-constant characteristics can be estimated
 - estimates are more efficient
- Cons:
 - if unobserved heterogeneity correlates with explanatory variables, coefficients are biased

The Within-Between Formulation

Using the within-between formulation, we can get 'FE estimates' from a RE model:

$$y_{it} = \beta_0 + \beta_1^{WE}(x_{1it} - \bar{x}_i) + \beta_1^{BE}\bar{x}_i + \gamma_1 z_{1i} + u_i + e_{it}$$

- This is a RE model, but any time-varying variable enters the model twice:
 - $\beta_1^{WE}(x_{1it} - \bar{x}_i)$ provides the within-effect of x_1
 - $\beta_1^{BE}\bar{x}_i$ provides the between effect of x_1
 - this is, the effect of x_1 is decomposed into its within- and between component, where the within-effect is equivalent to the FE estimator (Bell, Fairbrother, and Jones 2019)

The Within-Between Formulation

The within-between model is more flexible than RE or FE

- It keeps time-constant variables z_i in the model
- The RE model assumes that within- and between-effects are identical. The within-between model relaxes that assumption
- Testing $H_0 : \beta^{WE} - \beta^{BE} = 0$ is equivalent to asking whether the two components are identical
 - the Hausman test is a global test that compares all coefficients in the model
 - the within-between model allows to test single coefficients for equivalence of within and between effects

The Mundlak model

The within-between model goes back to Mundlak (1978), who proposed the following model:

$$y_{it} = \beta_0 + \beta_1^{WE} x_{1it} + \beta_1^{BWD} \bar{x}_i + \gamma_1 z_{1i} + u_i + e_{it}$$

, where β_1^{WE} is the within estimate, equivalent to the FE estimator, and β_1^{BWD} provides an estimate of the difference between within- and between effects ($\beta_1^{BE} = \beta_1^{WE} + \beta_1^{BWD}$)

- This model provides a direct test of the equivalence of within- and between effects: If β_1^{BWD} is significant, then within- and between-effects are different

Let's Estimate These Models in R

```
# Generate group-means and group-mean centered variables
wagepan_t <- wagepan %>% group_by(nr) %>%
  mutate(
    exper_m = mean(exper, na.rm = TRUE),
    union_m = mean(union, na.rm = TRUE)
  ) %>% ungroup() %>%
  mutate(
    exper_dm = exper-exper_m,
    union_dm = union-union_m
  )
# Declare data to be a panel data set (plm)
wagepan_t <- pdata.frame(wagepan_t, index = c("nr", "year"),
                        drop.index = TRUE)
# Estimate within-between and Mundlak model
wb      <- plm(lwage ~ black + hisp + educ + exper_m + exper_dm +
              union_m + union_dm, data=wagepan_t, model="random")
mundlak <- plm(lwage ~ black + hisp + educ + exper_m + exper +
              union_m + union, data=wagepan_t, model="random")
```

Table 12: Models of log hourly wages, wagepan data

	RE	FE	WB	Mundlak
(Intercept)	-0.075 (0.111)		0.241 (0.180)	0.241 (0.180)
black	-0.158*** (0.048)		-0.175*** (0.048)	-0.175*** (0.048)
hisp	0.015 (0.043)		0.007 (0.043)	0.007 (0.043)
educ	0.111*** (0.009)		0.096*** (0.011)	0.096*** (0.011)
exper	0.063*** (0.002)	0.064*** (0.002)		0.064*** (0.002)
union	0.113*** (0.018)	0.086*** (0.019)		0.086*** (0.019)
exper_m			0.036** (0.011)	-0.028* (0.011)
exper_dm			0.064*** (0.002)	
union_m			0.271*** (0.046)	0.186*** (0.050)
union_dm			0.086*** (0.019)	
s_idios	0.354		0.354	0.354
s_id	0.328		0.328	0.328
R ²	0.165	0.165	0.169	0.169
Adj. R ²	0.164	0.045	0.168	0.168
Num. obs.	4360	4360	4360	4360

The panelr Package

The within-between model is also implemented in the **panelr** package

```
# Declare wagepan data as panel data object
wagepan_pd <- panel_data(wagepan, id=nr, wave=year)
# Estimate model
wb_model <- wbm(lwage ~ exper + union |
                black + hisp + educ, data=wagepan_pd)
```

- the '|' in the formulation of the regression function separates variables to decompose (before the '|') from variables which are not decomposed (after the '|')

The panelr Package

```
##
## =====
##                               Model 1
## -----
## (Intercept)                0.241
##                          (0.180)
## `imean(exper)`              0.036 **
##                          (0.011)
## `imean(union)`              0.271 ***
##                          (0.046)
## exper                       0.064 ***
##                          (0.002)
## union                       0.086 ***
##                          (0.019)
## black                       -0.175 ***
##                          (0.048)
## hisp                         0.007
##                          (0.043)
## educ                        0.096 ***
##                          (0.011)
## -----
## AIC                         4505.975
## BIC                         4569.777
## Log Likelihood              -2242.987
## Num. obs.                   4360
## Num. groups: nr             545
## Var: nr (Intercept)         0.108
## Var: Residual               0.125
## =====
## *** p < 0.001; ** p < 0.01; * p < 0.05
```

When to Use Which Model? Additional Considerations

- Within-estimation has been introduced as the preferred estimator
- However, within-estimation requires within-unit variation
- Some variables simply do not change over time; others hardly change
 - know your data! Have a look at the ICC of each variable
 - if within-estimation identifies effects based on very small fractions of the data, it may not be a good strategy. Example:
 - we are interested the effect of electoral systems on voter turnout
 - say we have 26 countries, observed over 50 years. Sounds promising- doesn't it?
 - now, say only two countries changed their electoral system
 - FE could be estimated, but effects would be identified from these two cases

What About Time?

Should time be included as a variable in the model?

- Sometimes we have an explicit interest in describing developments over time
 - this is probably more often the case with pooled TSCS data than with individual-level panels.
 - pooled TSCS data can be representative of certain groups (e.g. all OECD countries, European countries, NATO members, etc.).
 - individual-level panel data is not ideally suited to describe social change over time because of panel attrition and changes in the population
- Time can also be an important 'control' variable
 - if x and y both follow a time trend, not controlling for time can falsely attribute part of the time trend to the effect of x

How to Specify Time in a Panel Model?

- It is a good idea to compare different models/parameterizations of time and then decide for one approach, which is used for the rest of the analysis
 - the most flexible modelling of time is including a dummy variables for each time point However, this is also the most demanding model with regard to the number of parameters
 - the simplest model is a linear model; it has just one parameter
 - adding polynomials (quadratic, cube, etc.) makes the function more flexible
- When you use LR tests, the most flexible model likely always 'wins'. Since we want to have parsimonious models, it is better to use a model fit indicator that penalizes complexity (e.g. AIC, BIC)

Random Slopes in Panel Models?

- The RE model in the traditional form includes only random intercepts
 - the RE model for panel data has been independently developed
 - it can be estimated without maximum likelihood, with generalized least squares, but it cannot include random slopes
- Does it make sense to include random slopes in panel models?
 - yes!
 - we can consider effect heterogeneity in panel models
 - we can also allow for individual time trends, which is something we will be looking at in the context of growth curve models

Summary: Analyzing Panel Data (Part II)

- The RE model uses within- and between-unit variation and can, thus, include time-constant variables
- It will, however, be biased if unobserved heterogeneity (u_i) is correlated with the variables in the model
- The Hausman test compares FE and RE and helps to pick the 'better' model
- The within-between or Mundlak formulations combine the virtues of FE and RE in one model
- Time has to be considered as a covariate

RE and RI - What's the Difference

The difference between panel and multilevel modeling lies in the way we use the hierarchical data structures to answer (different) research questions

- In panel data analysis u_i is called **unobserved heterogeneity**
 - It captures differences between individuals and appears to be a problem
 - This is why FE models get rid of it and RE models are only recommended when u_i is uncorrelated with the variables in the model
 - The main interest then lies in variables that vary over time (x_{it})
 - In sum, panel data is mostly used to aim for 'causal identification'

RE and RI - What's the Difference

The difference between panel and multilevel modeling lies in the way we use the hierarchical data structures to answer (different) research questions

- With multilevel modeling, we are very often interested in countries and u_j is called the **random intercept**
 - The random intercept captures the differences between level-2 clusters (i.e. countries) and it suddenly appears to be 'very interesting'
 - Multilevel modelers typically are interested in explaining differences between level-2 clusters (i.e. countries). This is, explaining the variation in u_j
 - Often we have a specific interest in context-level effects (z_j)
 - In sum, multilevel models are somewhat focused on the higher levels (i.e. country-level differences and effects)

Some Additional Topics

There are many topics specific to panel data analysis, which we could not cover. . .

- Autoregressive error terms
- First difference estimation
- Lagged variables and dynamic panel models

Andreß, Golsch, and Schmidt (2013) or Wooldridge (2010) are good sources for many additional topics

Exercise 10: Random Effects Models for Panel Data in R

- Open the PDF “Exercise 10”; it will walk you through an analysis task.

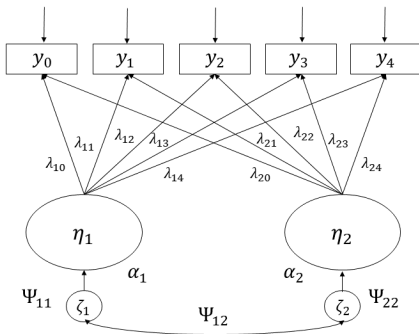
Growth Curve Models

- Growth curve models (GCMs) are used to model a dependent variable's development over time
 - these models are also called latent growth curve models, indicating that the average growth curve is viewed as a latent phenomenon
 - we aim to estimate the latent growth curve from the observed trajectories
 - growth curve modeling is well-suited when our main goal is to describe the development over time, as compared to testing the causal effect of a specific variable
- Examples:
 - How does public spending develop over time?
 - How does the size of work-related networks develop over time?
 - How does happiness develop over time in the life course?

A Growth Curve Model as a SEM

Latent growth curve modeling originated from structural equation modeling (SEM), for which you would have to use panel data in wide format

A latent growth curve model as a SEM



$$y_t = \lambda_{1t}\eta_1 + \lambda_{2t}\eta_2 + \varepsilon$$

$$\begin{aligned}\eta_1 &= \alpha_1 + \zeta_1 \\ \eta_2 &= \alpha_2 + \zeta_2\end{aligned}$$

A Latent Growth Curve Model as a Mixed Model

- But, as it turns out, it can also be estimated with mixed models
 - latent variables and random effects are essentially the same (Skrondal and Rabe-Hesketh 2004)
 - estimating the time trend with a random slope on it gives a latent growth curve model:

$$y_{it} = \beta_0 + \beta_1 t_{it} + u_{0i} + u_{1i} t_{it} + e_{it}$$

- Typically, such models will have t^2 as an additional variable (it is called growth **curve** and not growth line modeling):

$$y_{it} = \beta_0 + \beta_1 t_{it} + \beta_2 t_{it}^2 + u_{0i} + u_{1i} t_{it} + u_{2i} t_{it}^2 + e_{it}$$

- t should be coded to start with 0 so that the constant has a meaningful interpretation

Explaining Growth Curves

- GCM are often descriptive and simply describe the variance in intercepts and slopes over time, but they can include explanatory variables
- Let us write down the model in the separate equation approach and add a time-constant variable z_i :

- level-1 equation:

$$y_{it} = \beta_{0i} + \beta_{1i}t_{it} + \beta_{2i}t_{it}^2 + e_{it}$$

- level-2 equation intercept:

$$\beta_{0i} = \beta_0 + \gamma_{01}z_i + u_{0i}$$

- level-2 equation slope of t :

$$\beta_{1i} = \beta_1 + \gamma_{11}z_i + u_{1i}$$

- level-2 equation slope of t^2 :

$$\beta_{2i} = \beta_2 + \gamma_{12}z_i + u_{2i}$$

Explaining Growth Curves

- Putting everything back together:

$$y_{it} = \beta_0 + \beta_1 t_{it} + \beta_2 t_{it}^2 + \gamma_{01} z_i + \gamma_{11}(t_{it} z_i) + \gamma_{12}(t_{it}^2 z_i) + \\ u_{0i} + u_{1i} t_{it} + u_{2i} t_{it}^2 + e_{it}$$

- with the introduction of 'main effects' we explain variation in the intercepts of the growth curves
- explaining the slopes can be done by including 'cross-level' interactions between time and the variable of interest
- as the growth curve is a constant characteristic of each unit, it conceptually makes only sense to have time-constant variables as influences on them
- time-varying variables, however, can be included in the model, to partial out their effects

R Example: Estimating the Latent Growth Curve of Pigs

```
# Recode time to start with 0
pigs$t <- pigs$week-1
pigs$tsq <- pigs$t^2
# A linear and a quadratic growth curve model of pig weights
gcm1 <- lmer(formula=weight ~ 1 + t + (1 + t |pigID), data=pigs)
gcmq <- lmer(formula=weight ~ 1 + t + tsq + (1 + t + tsq |pigID), data=pigs)
```

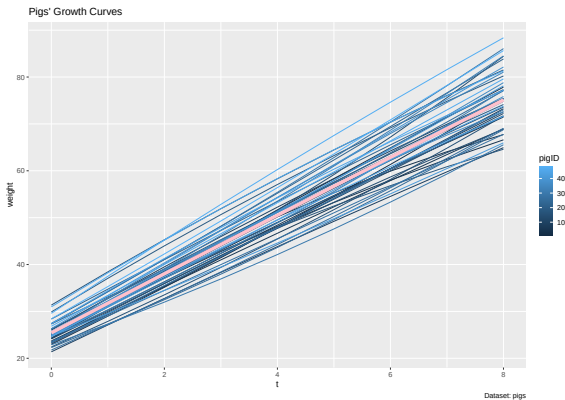
```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :  
## Model failed to converge with max|grad| = 0.00324505 (tol = 0.002, component 1)
```

Table 13: Linear and quadratic growth curve models of pig weight

	Linear	Quadratic
(Intercept)	25.57*** (0.40)	25.43*** (0.36)
t	6.21*** (0.09)	6.33*** (0.14)
tsq		−0.01 (0.02)
AIC	1752.87	1700.07
BIC	1777.28	1740.76
Log Likelihood	−870.44	−840.04
Num. obs.	432	432
Num. groups: pigID	48	48
Var: pigID (Intercept)	7.16	5.38
Var: pigID t	0.38	0.71
Cov: pigID (Intercept) t	0.28	0.89
Var: Residual	1.60	1.14
Var: pigID tsq		0.01
Cov: pigID (Intercept) tsq		−0.06
Cov: pigID t tsq		−0.06
*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$		

R Example: Visualizing Growth Curves

```
pigs$prediction = predict(gcmq) # prediction of individual lines
pigs$aprediction = predict(gcmq, re.form=NA) # prediction of average line
ggplot(pigs, aes(x=t, y=weight)) +
  geom_line(data=pigs, aes(x=t, y=prediction, group=pigID, color=pigID)) +
  geom_line(data=pigs, color="pink", linewidth=2, aes(x=t, y=aprediction)) +
  labs(title="Pigs' Growth Curves", caption="Dataset: pigs")
```



GCM in SEM vs. Multilevel Models

- Growth curve models can be specified in the SEM framework and in the multilevel framework
- We can formulate models that provide identical results in both frameworks
- With the SEM approach it is easier. . .
 - to specify error correlations, e.g. measurement errors over time correlate (but is also possible in mixed models)
 - to use a latent variable measurement model if y is latent (not possible in multilevel models)
- With the multilevel approach it is easier. . .
 - to include time-changing variables into the model
 - to set up and interpret the model

Exercise 11: Growth Curve Models in R

- Not today, tomorrow! Open the PDF “Exercise 11”; it will walk you through an analysis task with the garmit data.

Literature

- Agresti, Alan. 2012. *Categorical Data Analysis*. Vol. 792. John Wiley & Sons.
- Allison, Paul D. 1999. "Comparing Logit and Probit Coefficients Across Groups." *Sociological Methods & Research* 28 (2): 186–208.
- Andreß, Hans-Jürgen, Katrin Golsch, and Alexander W Schmidt. 2013. *Applied Panel Data Analysis for Economic and Social Surveys*. Springer Science & Business Media.
- Beck, Nathaniel. 2001. "Time-Series–Cross-Section Data: What Have We Learned in the Past Few Years?" *Annual Review of Political Science* 4 (1): 271–93.
- Bell, Andrew, Malcolm Fairbrother, and Kelvyn Jones. 2019. "Fixed and Random Effects Models: Making an Informed Choice." *Quality & Quantity* 53 (2): 1051–74.
- Breen, Richard, Kristian Bernt Karlson, and Anders Holm. 2021. "A Note on a Reformulation of the KHB Method." *Sociological Methods & Research* 50 (2): 901–12.
- Breusch, Trevor S, and Adrian R Pagan. 1980. "The Lagrange Multiplier Test and Its Applications to Model Specification in Econometrics." *The Review of Economic Studies* 47 (1): 239–53.
- Browne, William J, Harvey Goldstein, and Jon Rasbash. 2001. "Multiple Membership Multiple Classification (MMMC) Models." *Statistical Modelling* 1 (2): 103–24.

- Croissant, Yves, and Giovanni Millo. 2008. "Panel Data Econometrics in R: The plm Package." *Journal of Statistical Software* 27 (2): 1–43.
<https://doi.org/10.18637/jss.v027.i02>.
- Faraway, Julian J. 2016. *Extending the Linear Model with r: Generalized Linear, Mixed Effects and Nonparametric Regression Models*. Chapman; Hall/CRC.
- Garner, Catherine L, and Stephen W Raudenbush. 1991. "Neighborhood Effects on Educational Attainment: A Multilevel Analysis." *Sociology of Education*, 251–62.
- Garrett, Geoffrey, and Deborah Mitchell. 2001. "Globalization, Government Spending and Taxation in the OECD." *European Journal of Political Research* 39 (2): 145–77.
- Hausman, Jerry A. 1978. "Specification Tests in Econometrics." *Econometrica: Journal of the Econometric Society*, 1251–71.
- Johnson, David R, and Jian Wu. 2002. "An Empirical Test of Crisis, Social Selection, and Role Explanations of the Relationship Between Marital Disruption and Psychological Distress: A Pooled Time-Series Analysis of Four-Wave Panel Data." *Journal of Marriage and Family* 64 (1): 211–24.

- Karlson, Kristian Bernt, Anders Holm, and Richard Breen. 2012. "Comparing Regression Coefficients Between Same-Sample Nested Models Using Logit and Probit: A New Method." *Sociological Methodology* 42 (1): 286–313.
- Leckie, GB. 2012. "Cross-Classified Multilevel Models Using Stata: How Important Are Schools and Neighbourhoods for Students' Educational Attainment?" In *Hierarchical Linear Modeling: Guide and Applications*, 311–32. SAGE Publications Ltd.
- Long, Jacob A. 2020. *Panelr: Regression Models and Utilities for Repeated Measures and Panel Data*. <https://cran.r-project.org/package=panelr>.
- Mood, Carina. 2010. "Logistic Regression: Why We Cannot Do What We Think We Can Do, and What We Can Do about It." *European Sociological Review* 26 (1): 67–82.
- Mundlak, Yair. 1978. "On the Pooling of Time Series and Cross Section Data." *Econometrica: Journal of the Econometric Society*, 69–85.
- Rasbash, Jon, and William Browne. 2001. "Modelling Non-Hierarchical Structures," [https://citeseerx.ist.psu.edu/document?repid=rep1 & type=pdf & doi=1bd087e2c7850f3acd4a611936e28894e0aa07b0](https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=1bd087e2c7850f3acd4a611936e28894e0aa07b0).

- Schmidt-Catran, Alexander W, Malcolm Fairbrother, and Hans-Jürgen Andreß. 2019. "Multilevel Models for the Analysis of Comparative Survey Data: Common Problems and Some Solutions." *KZfSS Kölner Zeitschrift für Soziologie Und Sozialpsychologie* 71 (1): 99–128.
- Shi, Yuying, Walter Leite, and James Algina. 2010. "The Impact of Omitting the Interaction Between Crossed Factors in Cross-Classified Random Effects Modelling." *British Journal of Mathematical and Statistical Psychology* 63 (1): 1–15.
- Skrondal, Anders, and Sophia Rabe-Hesketh. 2004. *Generalized Latent Variable Modeling: Multilevel, Longitudinal, and Structural Equation Models*. Chapman; Hall/CRC.
- Wooldridge, Jeffrey M. 2010. *Econometric Analysis of Cross Section and Panel Data*. MIT press.
- . 2015. *Introductory Econometrics: A Modern Approach*. Cengage learning.